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DCDA-Net: A depthwise convolution dilated attention network with temporal weighted loss for seizure prediction

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ABSTRACT

The sudden and unforeseeable nature of seizures poses serious threats to patients' safety and quality of daily life, making the development of highly accurate and reliable seizure prediction methods of significant clinical importance. To this end, this study proposed a seizure prediction model by integrating depthwise convolution and a novel dilated attention mechanism, named DCDA-Net. Depthwise convolution was first employed to extract features from each EEG channel independently, preserving spatial channel information for subsequent seizure evolution analysis. Innovatively combining dilated convolution with self-attention mechanisms, a dilated attention module was devised to efficiently capture long-range temporal dependencies. Furthermore, a temporal weighted loss function was introduced to significantly enhance the model's ability to discriminate critical preictal temporal features. Validation results on the CHB-MIT and SH-SDU clinical databases demonstrated that the model achieved a segment-based accuracy of 97.56% and an event-based sensitivity of 97.69% with a false prediction rate of 0.0232/hour on the CHB-MIT database, while attaining a segment-based accuracy of 96.93% and an event-based sensitivity of 96.05% on the SH-SDU database. Additionally, channel importance analysis confirmed significant spatial consistency with the actual epileptogenic zones. This study offered an efficient solution for patient-specific seizure prediction, with substantial potential for further clinical assessment.

1. Introduction

Epilepsy is a common and severe chronic neurological disorder that affects more than 50 million individuals worldwide (Shan et al., 2025). Epileptic seizures happen abruptly and are unpredictable, which not only increases the risk of accidental injuries and sudden unexpected death but also imposes persistent anxiety and fear on patients, thereby severely impairing their quality of life (Zhao et al., 2024). Current therapeutic strategies primarily rely on antiepileptic drugs and neurostimulation techniques, aiming to suppress seizure onset or to reduce seizure frequency. In contrast, the effectiveness of these passive controls has been limited. Instead, the ability to accurately predict impending seizures and provide timely warnings or preventive interventions holds transformative potential, and could potentially improve patients' quality of life and long-term outcomes (Cui et al., 2024; Shafiezadeh et al., 2024).

Electroencephalography (EEG), a non-invasive method for monitoring cerebral electrical activity, has become an indispensable tool in epilepsy diagnosis and seizure prediction due to its high temporal resolution and ability to capture synchronous neuronal discharges (Dong et al., 2024a). In seizure prediction research, a key task is to identify the preictal state within EEG signals reliably. This state is hypothesized to exist as a transitional phase between normal background activity (interictal) and clinical seizure onset (ictal), characterized by neural patterns that precede seizure occurrence(Qi et al., 2026; Wang et al., 2026; Zhang et al., 2020). Therefore, the essence of seizure prediction

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is to accurately detect these transient and subtle preictal patterns from long, complex, and noisy interictal EEG recordings using computational and analytical methods, thereby enabling timely warning before seizure onset (Liu et al., 2023).

As a powerful paradigm of machine learning, deep learning has recently demonstrated tremendous potential and achieved remarkable progress in seizure prediction and other fields (Chen et al., 2025; Jahan et al., 2023). Deep learning models consist of multi-layer neural networks that can automatically learn highly abstract and discriminative spatiotemporal representations from raw or minimally preprocessed EEG signals in an end-to-end manner (Kumar et al., 2026; Li et al., 2024). This approach eliminates the reliance on traditional handcrafted feature engineering, thus overcoming its inherent limitations of complexity and poor generalization (Varnosfaderani et al., 2024). With such strong representation learning capability, deep learning models can more effectively capture the subtle and evolving dynamical changes within the preictal phase, offering a promising pathway toward accurate and robust seizure prediction.

Despite these advancements, EEG-based seizure prediction still faces several major challenges. First, scalp EEG signals are inherently complex, with different frequency bands (e.g., δ, θ, α, β, γ) corresponding to distinct physiological or pathological processes (Iyer et al., 2022; Miao et al., 2023). These components collectively form a dynamic and mixed background of brain activity, making it difficult to isolate seizure-related frequency patterns that are critical for prediction. Second, both preictal and interictal EEGs represent normal brain activity between seizures. Hence, it is even more challenging to distinguish the preictal state from the interictal state than from the ictal periods (Rasheed et al., 2021). Furthermore, seizure-related EEG waveforms usually originate from a few channels in the epileptogenic zone and gradually propagate to neighboring brain regions over time, leading to complex changes in the contribution of different channels during seizure evolution. Currently, most deep learning-based seizure detection methods mainly focus on seizure-state classification, while lacking in-depth exploration of epileptogenic brain region information (Jemal et al., 2022; Ren et al., 2026). Some machine learning approaches attempted to identify seizure-related regions by establishing the relationship between model decisions and EEG channels. Birjandtalab et al. (2017) proposed a seizure detection framework combining limited-channel EEG with nonlinear dimensionality reduction, and employed random forest to identify the EEG channels that contributed most to seizure events. Li et al. (2021) developed a seizure detection method integrating CSP and SVM, in which CSP spatial-filter weights were employed to identify EEG channels involved in seizure onset. However, these methods mainly focus on post-hoc interpretation of output results and cannot provide information on the evolution of seizure activity.

To address the aforementioned challenges, this study proposes a deep learning framework, DCDA-Net, coupled with temporal weighted loss for interpretable seizure prediction. By integrating depthwise convolution, dilated self-attention, and temporal weighting mechanism, DCDA-Net can dynamically focus on the key EEG channels while capturing EEG long-range dependencies. This design effectively addresses the challenges of key-channel identification, long-term dependency modeling, and preictal state discrimination, enhancing both the accuracy and interpretability of seizure prediction. The main contributions of this work are summarized as follows:

1) **Dilated Attention Mechanism for Multi-scale Long-range Dependency Modeling:** To overcome the difficulty of capturing multi-scale long-range dependencies in multi-channel EEG signals, we propose a novel dilated attention mechanism that integrates the multi-scale modeling capability of dilated convolutions with the global dependency modeling of self-attention. This mechanism significantly expands the receptive field and performs attention over sparsely sampled temporal positions with multiple dilation rates, enabling the model to jointly analyze local fluctuations and global trends for precise seizure prediction.
2) **Temporal Weighted Loss for Preictal-interictal Discrimination:** To address the challenge of distinguishing preictal from interictal states, we introduce a temporal weighted loss function during training. This function is able to dynamically assign higher weights to time segments closer to the clinical seizure onset, guiding the model to focus on the most critical prediction window. This approach enhances both the timeliness and sensitivity of prediction while effectively reducing false alarms.
3) **Quantitative Analysis for Key Channel Identification:** To deal with the issue of identifying critical EEG channels, we further perform a channel-wise quantitative analysis to evaluate the contributions of different EEG channels to seizure prediction based on the L2 norm of the corresponding depthwise convolutional kernels, therefore providing critical insights into the evolution of seizure precursors.
4) **Frequency-aware Interpretability Analysis:** To enhance model interpretability and explore neurophysiological relevance, we employ Gradient-weighted Class Activation Mapping (Grad-CAM) to perform a frequency-aware analysis. The resulting heatmaps reveal the model's decision-making process across different frequency bands, bridging the gap between deep learning predictions and clinical interpretation.

We evaluated the performance of our model using the public CHB-MIT database and the clinical SH-SDU database. Experimental results demonstrate that the proposed deep learning framework outperforms existing mainstream methods in key metrics such as prediction sensitivity and false prediction rate. Most importantly, we have for the first time systematically revealed key spatial features of the preictal phase from a data-driven perspective, enabling efficient localization of epileptogenic channels. Our work offers an advanced technical pathway for seizure prediction and lays a methodological foundation for its clinical translation and interpretation of seizure evolution mechanisms.

The remainder of this paper is organized as follows: Section 2 reviews related work. Section 3 provides a detailed description of the proposed method. Section 4 presents the experimental results. Section 5 discusses the findings and compares them with existing studies. Finally, Section 6 concludes this work.

2. Related work

Seizure prediction research has evolved over many years from using traditional machine learning approaches based on handcrafted features to modern end-to-end deep learning methods. The following sections provide a detailed overview of these two stages.

2.1. Machine learning methods with manual features

Early studies primarily relied on handcrafted features combined with shallow machine learning models for EEG classification. These features quantify linear and nonlinear dynamics, including time-domain features (mean and variance), frequency-domain features (band power and spectral edge frequency), and nonlinear features (entropy and fractal dimension) (Ruiz de Miras et al., 2023; Tibdewal et al., 2017). They are typically fed into classifiers such as Support Vector Machines (SVMs), Random Forests (RFs), or k-Nearest Neighbors (KNNs) to distinguish interictal from preictal EEG (Donos et al., 2015; Murugavel & Ramakrishnan, 2015; Rajalakshmi & Sridhar, 2024). For example, Savadkoohi et al. (2020) extracted statistical features from δ, θ, α, β, and γ bands using Butterworth filtering, Fourier transform, and wavelet transform, followed by feature selection and classification with KNN and SVM. Wang et al. (2019) used db4 wavelet packet decomposition to derive sub-band energy ratios and entropy-based features and classified with RF. They achieved an accuracy of 85.20%. Qureshi and Kaleem

(2022) employed db6 wavelet packet decomposition to extract energy, standard deviation, signal peaks, and spectral derivatives from δ, θ, α, and β bands, and used KNN, SVM, and shallow neural networks for classification. Among them, the SVM achieved the best performance with 94.94% accuracy and 97.43% sensitivity. While foundational, these methods were limited by their heavy reliance on handcrafted feature design and domain knowledge, and often lacked generalization to capture individual variability and complex seizure dynamics.

2.2. Deep learning methods

In recent years, deep learning has transformed seizure prediction by automatically learning discriminative high-level features from raw or minimally preprocessed EEG in an end-to-end manner, avoiding handcrafted feature engineering (Zhang et al., 2024). Convolutional neural networks (CNNs) capture the spatial relationships across EEG channels and local time-frequency patterns, either by converting multi-channel EEG into two-dimensional time-frequency maps or using one-dimensional CNNs applied directly to time series (Li et al., 2022; Priya Prathaban & Balasubramanian, 2021). Furthermore, long short-term memory networks (LSTMs) and their variants model long-range temporal dependencies and they can effectively identify dynamic changes from interictal to preictal stages. Transformers, leveraging self-attention mechanisms, have also been applied to capture global dependencies in long EEG sequences (Dong et al., 2025; Wu et al., 2023).

Wang et al. (2024b) proposed a dual CNN–LSTM framework with a channel-reordering module. The module calculates the classification F1 score for each iEEG channel and ranks the channels accordingly. The model then repeats the classification using progressively larger subsets of the ranked channels to identify the optimal channel combination. It was reported that this approach could reduce computational cost while improving prediction performance. Ji et al. (2024) combined graph attention networks (GAT) with temporal convolutional networks (TCN) to extract spatial and temporal features, achieving satisfactory results. Bhattacharya et al. (2021) developed an end-to-end framework combining short-time Fourier transform features with a deep Transformer, achieving an average sensitivity of 98.46% on the CHB-MIT database.

3. Materials and methods

Fig. 1 illustrates the overall framework of the proposed automated seizure prediction method. It primarily includes signal acquisition, acquisition of different frequency bands, the DCDA-Net, and the loss functions used during training and testing. Detailed descriptions of each component will be provided in subsequent sections.

3.1. EEG database

In this study, the CHB-MIT scalp EEG database, collected from Boston Children's Hospital, was used (Shoeb, 2009). The EEG signals in this database were sampled at 256 Hz with a resolution of 16 bits and recorded according to the international 10-20 system. The database includes 24 cases from 23 patients, each containing multiple continuous EEG recordings. In total, it encompasses approximately 980 hours of recordings and 198 seizure events annotated by clinical experts. To ensure consistent channel coverage across patients, 18 common channels were selected for analysis: FP1-F7, F7-T7, T7-P7, P7-O1, FP1-F3, F3-C3, C3-P3, P3-O1, FP2-F4, F4-C4, C4-P4, P4-O2, FP2-F8, F8-T8, T8-P8, P8-O2, FZ-CZ, and CZ-PZ.

The second database used in this study is the SH-SDU database, collected at the Second Hospital of Shandong University. This database includes long-term scalp EEG recordings from 19 patients sampled at 500 Hz. Unlike the CHB-MIT database, the SH-SDU patients are mostly adults and exhibit a higher frequency of seizures during EEG recording. All patients had 18 EEG recording channels: Fp1, Fp2, F3, F4, C3, C4, P3, P4, O1, O2, F7, F8, T3, T4, T5, T6, A1, and A2, where A1 and A2 are ear electrodes. In this study, the A1 and A2 channels were excluded, and the remaining 16 channels were used for analysis. The SH-SDU database contains 135 seizure events, with a total recording duration of approximately 286.83 hours. We downsampled the data to 256 Hz to match the sampling frequency of the CHB-MIT database. Detailed information for the CHB-MIT and SH-SDU databases is listed in Table 1 and Table 2, respectively.

For seizure prediction tasks, it is essential to clearly define the preictal, postictal, and interictal periods. Following previous studies (Bou Assi et al., 2017; Dong et al., 2024b), we defined the 30 minutes preceding seizure onset as the preictal period, the 5 minutes following seizure termination as the postictal period, and the remaining time as

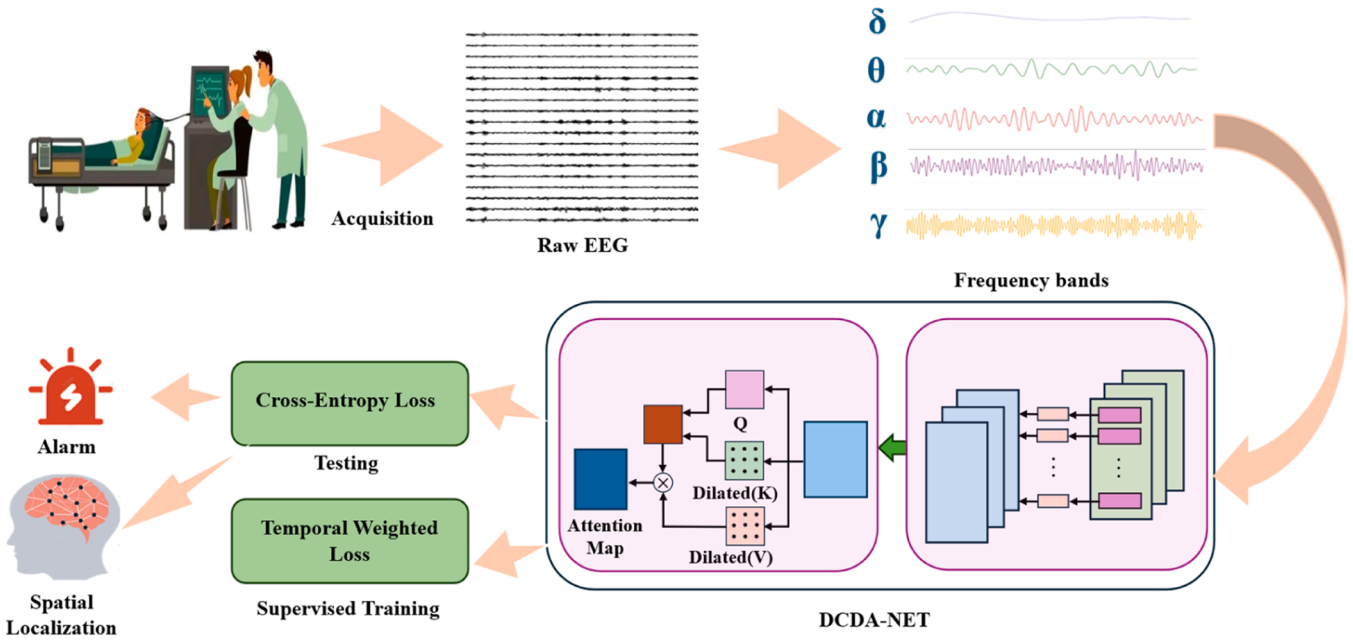


Fig. 1. Overall framework diagram of the proposed seizure prediction method.

Table 1
Specific information of the CHB-MIT database.

Patient	Gender	Age	Number of seizures	Number of used seizures	Duration of recordings (h)
1	F	11	7	7	40.55
2	M	11	3	3	35.27
3	F	14	7	6	38.00
4	M	22	4	4	156.07
5	F	7	5	5	39.00
6	F	1.5	10	9	66.74
7	F	14.5	3	3	67.05
8	M	3.5	5	5	20.01
9	F	10	4	4	67.87
10	M	3	7	6	50.02
11	F	12	3	0	34.79
12	F	2	40	7	20.69
13	F	3	12	4	33.00
14	F	9	8	6	26.00
15	M	16	20	9	40.01
16	F	7	10	5	19.00
17	F	12	3	3	21.01
18	F	18	6	4	35.63
19	F	19	3	2	29.93
20	F	6	8	4	27.60
21	F	13	4	4	32.83
22	F	9	3	2	31.00
23	F	6	7	4	26.56
24	-	-	16	0	21.30
Total	-	-	198	106	979.93

Table 2
Specific information of the SH-SDU database.

Patient	Gender	Age	Number of seizures	Number of used seizures	Duration of recordings (h)
1	F	28	10	5	12.00
2	F	28	11	6	23.74
3	M	53	7	5	11.92
4	M	79	32	5	12.00
5	M	8	10	2	6.00
6	F	38	3	3	6.00
7	M	34	2	2	6.00
8	M	61	10	6	16.04
9	M	78	4	4	15.79
10	M	71	3	3	12.00
11	M	64	15	6	18.00
12	M	66	4	2	6.00
13	F	17	2	2	19.38
14	M	25	3	3	17.95
15	F	37	7	6	44.63
16	F	38	3	3	17.38
17	M	14	3	3	8.00
18	F	37	4	4	17.50
19	M	61	2	2	16.50
Total	-	-	135	72	286.83

the interictal period (Fig. 2). Only seizure events meeting the inclusion criteria were retained for this study, with the detailed selection procedure described in the data preprocessing section.

3.2. Preprocessing

In the EEG database, some adjacent seizure events have time intervals of less than 30 minutes, which do not satisfy the definition of the preictal period (30 minutes) adopted in this study. Therefore, only adjacent seizure events with intervals greater than 30 minutes were selected for analysis in this experiment. Based on these criteria, 106 seizure events from the CHB-MIT database and 72 from the SH-SDU database were selected for subsequent analysis. This study utilized a sliding window of 4 seconds to segment both the 30-minute preictal data and the randomly selected interictal EEG data and formed the "Sample Set A" for subsequent analysis. Sample Set A has dimensions of (N, C, 1024), where N is the number of samples, C is the number of channels, with 1024 data points per sample.

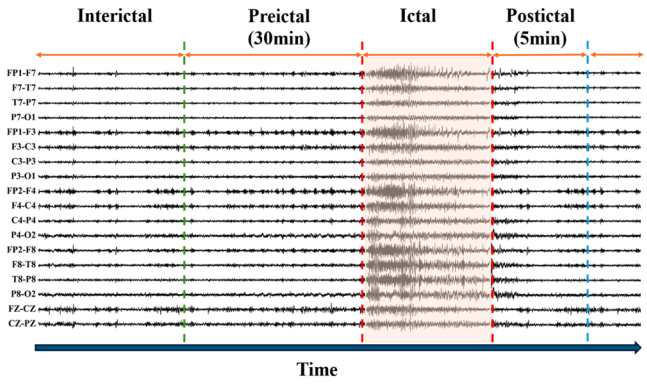


Fig. 2. 18-channel EEG, definitions of preictal, ictal, postictal, and interictal periods.

EEG signals can be divided into several frequency bands, such as δ, θ, α, β, and γ bands, which are closely associated with various neural activities. To remove power line interference, we employed notch filters: a 60 Hz notch filter for CHB-MIT database and a 50 Hz notch filter for SH-SDU database. To investigate the potential role of specific frequency bands during the preictal phase, we applied a 5th-order Butterworth bandpass filter to each channel in Sample Set A to isolate the δ, θ, α, β, and γ bands, respectively (Fig. 3). Subsequently, the five band signals extracted from each channel were concatenated sequentially along the channel dimension to construct a new multi-dimensional "Sample Set B". Sample Set B has dimensions of (N, $C \times 5$, 1024), where N is the number of samples, $C \times 5$ represents 5 frequency bands from each channel, and 1024 is the number of data points per sample.

3.3. DCDA-Net

The proposed DCDA-Net consists of two primary parts: the depthwise convolution module and the dilated attention module. Each of these modules is introduced in detail below, providing their specific roles and

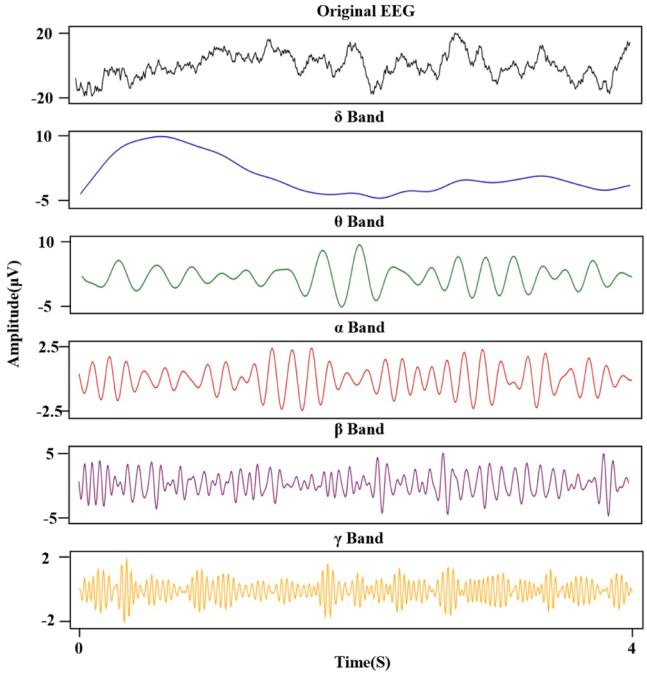


Fig. 3. The δ, θ, α, β, and γ waves obtained after filtering.

contributions to the overall framework.

3.3.1. Depthwise convolution module

Depthwise convolution is a core component of the depthwise separable convolution architecture, and its operation differs fundamentally from that of standard convolution (Liu et al., 2026; Wang et al., 2024a). In standard convolution, each kernel operates across all input channels simultaneously to perform cross-channel feature integration. In contrast, the depthwise convolution follows a channel-wise one-to-one mapping principle: each kernel operates exclusively on a single input channel, and each input channel is processed by only one independent kernel. When the number of input channels is *C*, *C* independent convolutional kernels are used to perform spatial filtering within their respective channels. The depthwise convolution modular architecture ensures its adaptability to varying numbers of EEG channels in practical applications.

Since this process is conducted entirely independently across input channels, depthwise convolution inherently involves no cross-channel information interaction. Although this characteristic prevents it from directly leveraging feature correlations from different channels at the same spatial locations, it offers a critical advantage in that it compels the network to distinctly separate the unique spatial filtering patterns of each channel during the learning process. Such explicit modeling of channel-specific features facilitates the subsequent identification of key frequency bands and spatial locations associated with seizures.

The architecture of the proposed depthwise convolution module is illustrated in Fig. 4. Taking the CHB-MIT database as an example, the preprocessed 90-dimensional input data, formed by sequentially arranging signals from different frequency bands (18 EEG channels × 5 frequency bands), passes through three feature extraction stages. Each stage consists of a depthwise convolution layer, a batch normalization layer, a ReLU activation function, and a pooling layer. While the first two stages employ standard pooling, the third stage applies adaptive pooling to map the feature tensor to a fixed dimension. All depthwise convolution layers use a kernel size of 3 and a stride of 1. Through progressive transformations across the three stages, the input data with an initial size of (90, 1024) are ultimately mapped to a feature matrix of size (1, 90, 90). Here, the output tensor has a size of (1,90,90), where the second dimension corresponds to the 90 channel-band units formed by concatenating 18 EEG channels and 5 frequency bands, while the third dimension represents the compressed temporal feature representation with a fixed length of 90.

3.3.2. Dilated attention module

In EEG-based seizure prediction, capturing multi-scale temporal features is essential, as these features encompass both transient spike-and-wave discharges and sustained rhythmic patterns. Traditional models typically expand the receptive field by stacking layers, but this approach increases computational complexity and often leads to vanishing gradient issues. To address these limitations, dilated convolution has been introduced as a core feature extraction method (Bai, 2018). By introducing dilation rates, it achieves an exponential expansion of the receptive field without increasing the number of parameters and has demonstrated strong effectiveness in seizure detection tasks (Wang et al., 2025). Meanwhile, self-attention mechanisms, with their powerful capability for modeling global dependencies, offer a new perspective for capturing long-range relationships in EEG data (Vaswani et al., 2017). Variants such as the Vision Transformer, Swin Transformer, and Informer further enhance this capability and have shown outstanding performance in processing long-sequence EEG signals (Dosovitskiy, 2020; Liu et al., 2021; Zhou et al., 2021). Leveraging the complementary strengths of dilated convolution and self-attention, we proposed a dilated attention unit that restricts attention computation to dilation-defined local receptive fields, thereby effectively capturing broad contextual information while maintaining computational efficiency.

The core part of the proposed dilated attention module is the window dilated attention unit. For an input tensor $X \in R^{B\times C\times H\times W}$, Query, Key, and Value features are first generated through linear projections:

$$Q = W_q X \tag{1}$$

$$K = W_k X \tag{2}$$

$$V = W_v X \tag{3}$$

where W represents a learnable parameter. Q is Query, K is Key and V is Value.

In standard local attention, the attention range for each pixel position (i,j) is typically confined to a $k \times k$ window. In the proposed Dilate-Attention mechanism, we introduce a dilation rate r, and select sampling points in the unfolding operation according to the following scheme:

$$\Omega^{k}_{(i,j)} = \left\{ (i + r\cdot V_i, j + r\cdot V_j) \,|\, V_i, V_j \in [\,-\lfloor r/2 \rfloor, \lfloor r/2 \rfloor] \right\} \tag{4}$$

where r denotes the dilation factor. When $r > 1$, the spacing between sampling positions within the window is expanded, enabling each attention head to capture information interactions across broader spatial ranges. This process is implemented via the Unfold operation and can be formulated as follows:

$$K_{unfold} = Unfold(K; k, r) \tag{5}$$

$$V_{unfold} = Unfold(V; k, r) \tag{6}$$

where $Unfold(\cdot)$ denotes the unfolding operation with kernel size k and dilation rate r. Subsequently, the local dilated attention weights are computed for each position as follows:

$$A_{(i,j)} = Softmax\left(\frac{Q_{(i,j)} K^{T}_{unfold(i,j)}}{\sqrt{d_k}} \right) \tag{7}$$

where $\sqrt{d_k}$ represents the scaling factor. The final attention output is:

$$Y_{(i,j)} = A_{(i,j)} V_{unfold(i,j)} \tag{8}$$

To further achieve multi-scale feature fusion, this study proposes a multi-head window dilated attention module, as illustrated in Fig. 5. This module contains multiple parallel dilated attention branches, with each branch corresponding to a distinct dilation rate r to capture spatial

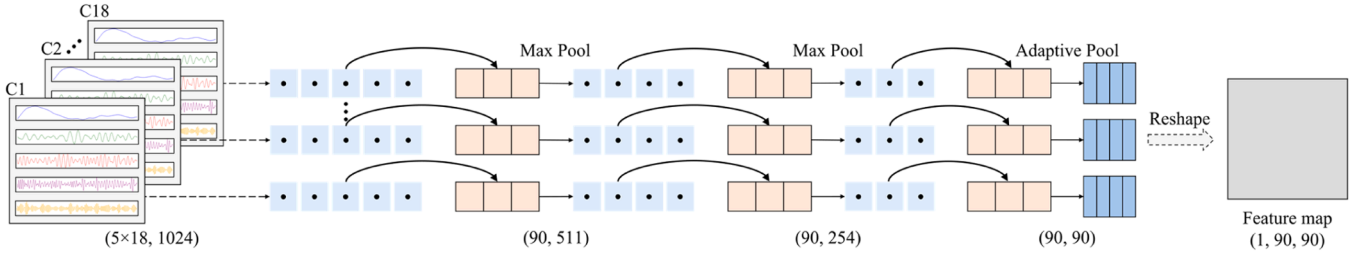


Fig. 4. The specific structure of the depthwise convolution module.

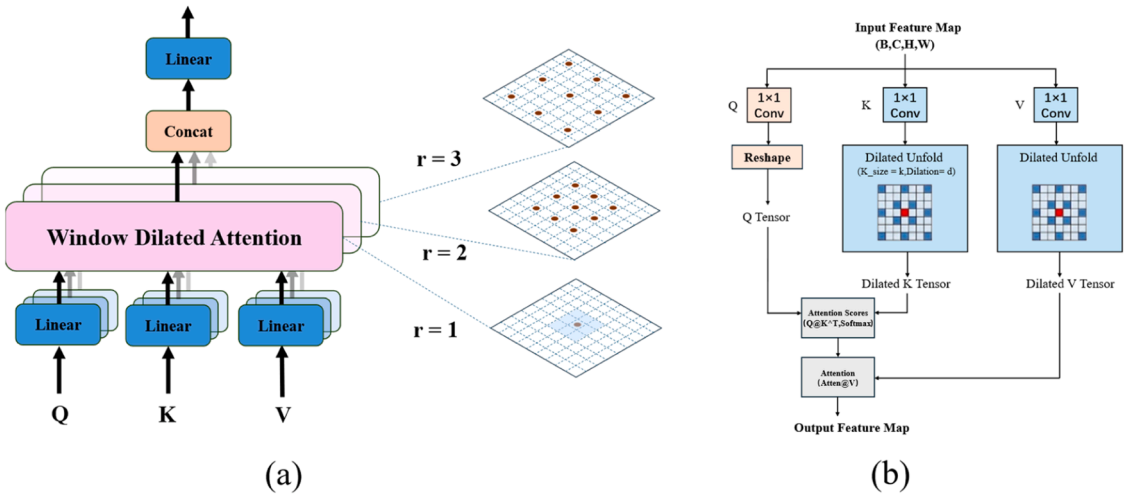


Fig. 5. The multi-head window dilated attention unit, with 3 heads and dilation factors r of 1, 2, and 3. (a) The specific structure of the multi-head window dilated attention unit. (b) The calculation process of attention.

contexts at different scales. The specific computation is as follows:

$$Y_i = DilateAttention(Q_i, K_i, V_i, r_i) \tag{9}$$

Finally, features from different dilation scales are concatenated and fused as follows:

$$Y = Linear(Concat(Y_1, Y_2, \ldots, Y_n)) \tag{10}$$

Fig. 6 illustrates the detailed structure of the dilated attention module, which is similar to the Swin Transformer architecture and can be divided into four main stages. However, the key difference is that the Swin Transformer blocks are replaced with dilated attention blocks. A feature matrix of size (1, 90, 90) first passes through the Patch Embedding layer, which is primarily composed of convolutional neural networks. The main function of this layer is to convert the input data from the original pixel space into the embedding space, preparing it for subsequent model processing. The embedding dimension is 36. In addition, the Linear Embedding layer, also implemented using convolutional neural networks, further transforms the feature dimensions of the model output. The dilated attention block mainly consists of a LayerNorm layer, a dilated attention layer, and a Multilayer Perceptron (MLP), along with residual connections, as shown in Fig. 6(b). After the first stage, the feature matrix of size (1, 90, 90) is transformed into a feature map of size (36, 23, 23). Subsequently, after each stage, the first dimension (channel dimension) doubles, while the second and third dimensions are halved. After four stages, the output feature matrix has a size of (288, 3, 3). Each stage comprises 2 layers. The number of attention heads in the Dilated Attention is 3, and the dilation rates are set at 1, 2, and 3, respectively. The dilated attention module achieves channel-wise information fusion by computing attention across channels, enabling the model to capture long-range temporal and inter-channel dependencies. Finally, the resulting feature matrix is flattened and passes through an MLP to produce the final prediction output.

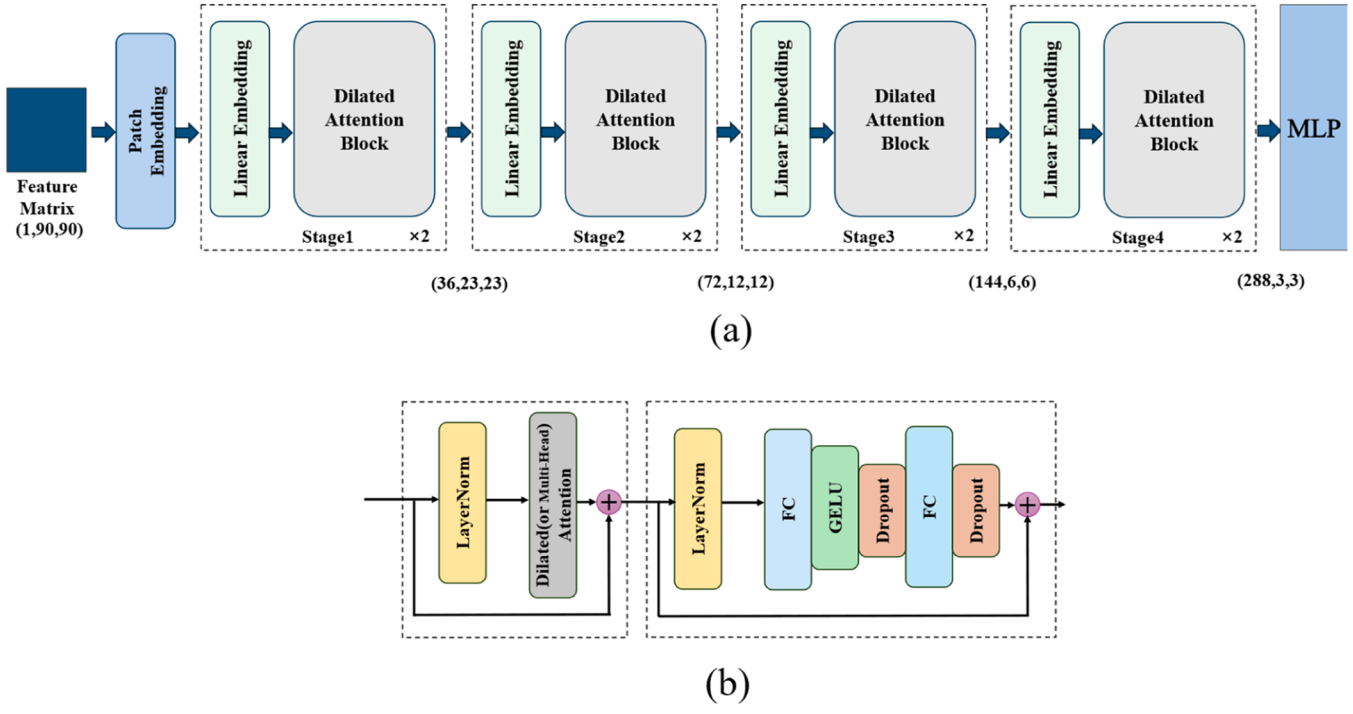


Fig. 6. The specific structure of the dilated attention module. (a) The four stages and their specific connections. (b) A dilated attention block.

The proposed Dilated Attention Module is not a simple serial or parallel combination of dilated convolution and self-attention. Instead, it incorporates the idea of dilated sampling into the self-attention computation process, thereby introducing an enhanced self-attention mechanism. Compared with conventional self-attention, its innovations are mainly reflected in the following two aspects.

(1) Design of a novel dilated self-attention mechanism: The proposed method directly embeds the dilated sampling mechanism into the self-attention computation, resulting in the Dilated Attention module. Specifically, during the generation of the Key and Value matrices, a Dilated Unfold operation is introduced. By introducing a configurable dilation rate, the attention computation is transformed from conventional dense sampling into a local receptive field modeling process based on dilated sampling. Compared with traditional local window attention, this approach can effectively enlarge the receptive field without increasing computational complexity, enabling the model to learn richer long-range contextual dependencies with fewer network layers, thereby enhancing its ability to model multi-scale spatiotemporal features.
(2) Construction of a hierarchical dilated attention architecture: Based on the hierarchical architecture of the classic Swin Transformer, the proposed method replaces the original window attention modules with the proposed Dilated Attention module, forming a hierarchical dilated attention framework. While retaining the advantages of Swin Transformer in progressive feature extraction, multi-stage hierarchical modeling, and multi-scale feature representation, the proposed architecture enables long-range information interaction and cross-scale feature fusion through the collaboration of dilated attention modules with different dilation rates. As the network depth increases, attention features at different scales continuously interact and aggregate across both spatial and channel dimensions, forming richer multi-scale long-range dependencies. Consequently, the proposed model provides more effective representations of complex EEG dynamics and further improves seizure prediction performance.

3.4. Temporal weighted loss

In seizure prediction tasks, EEG samples from different time points contribute unequally to prediction outcomes. Typically, signal segments closer to the clinical seizure onset contain more pronounced preictal characteristics and carry greater pathological significance, whereas signals farther from the onset often exhibit physiological patterns similar to normal EEG or only display weak and ambiguous pre-seizure patterns. The traditional cross-entropy loss function assigns equal weight to all samples during training, failing to capture the inherent temporal sensitivity of seizure prediction tasks, thereby limiting the model's ability to focus on critical preictal features (Rao et al., 2024).

To address the aforementioned limitations, this study proposes a temporal weighted loss function designed to effectively model the importance variation of samples across different pre-seizure time intervals. The method introduces an adjustable temporal decay coefficient ρ, which assigns differentiated weights to pre-seizure samples based on their temporal proximity to the seizure onset, thereby enabling the model to focus more intensively on critical periods immediately preceding seizure occurrence during training. Specifically, each pre-seizure segment is partitioned into N equidistant temporal samples. Each sample is assigned a temporal index t_i, where $t_i = 0$ denotes the segment closest to seizure onset. To quantify the influence of temporal proximity on sample importance, the temporal weight is defined as follows:

$$w_i = \frac{1}{1 + e^{\rho t_i}} \tag{11}$$

where $\rho > 0$ is the temporal decay coefficient, t_i represents the relative time to seizure onset (a larger value indicates an earlier segment). The range of t_i is from 0 to 1800, where the first sample is assigned a value of 1800, and the last sample is assigned a value of 0. Hence,$w_i$denotes the weight assigned to each sample, with larger values indicating closer proximity to seizure onset, thereby requiring greater attention from the model. The function is smooth and differentiable throughout its domain. The proposed temporal weighted loss function is defined as follows:

$$L_{TWLoss} = -\frac{1}{N}\sum_{i=1}^{N} w_i \sum_{c=1}^{C} y_{i,c}\log\left(p_{i,c}\right) \tag{12}$$

where N represents the total number of samples, C denotes the number of classes, $y_{i,c}$indicates the ground truth label of sample i for class$c$, and $p_{i,c}$represents the predicted probability of sample i belonging to class$c$.

In our experiments, ρ was set to 0.001 and w_i was fixed at 0.2 for interictal samples. The loss function first computes the standard cross-entropy loss for each sample and then performs a weighted summation based on the temporal weights to achieve temporally sensitive optimization. This approach effectively enhances the model's discriminative capability and temporal awareness in seizure prediction.

4. Results

4.1. Experiment settings and performance evaluation

The experiments were implemented using PyTorch 2.8.0 and Python 3.13.7 on a computer configured with an Intel (R) Core (TM) i9-14900HX CPU, 32 GB RAM and an NVIDIA GeForce RTX 5070 GPU. During the training phase, the Adam optimizer was employed with a learning rate of 0.0001, a batch size of 128, and a maximum of 50 iterations. Training and structural parameters (e.g., learning rate, batch

Table 3
Experimental settings of the proposed DCDA-Net.

Category	Layer	Parameters
Depthwise convolution module	Depthwise Conv1d-1	(90, 90, 3, 1, 1, 90)
	BatchNorm	(90)
	ReLU	-
	Max Pooling	(2)
	Depthwise Conv1d-2	(90, 90, 3, 1, 1, 90)
	BatchNorm	(90)
	ReLU	-
	Max Pooling	(2)
	Depthwise Conv1d-3	(90, 90, 3, 1, 1, 90)
	BatchNorm	(90)
	ReLU	-
	Adaptive Pooling	(90)
Dilated attention module	Patch Embedding	(36)
	Stage 1 Dilated Attention Blocks	(2, 3, (1, 2, 3))
	Stage 2 Dilated Attention Blocks	(2, 6, (1, 2, 3))
	Stage 3 Dilated Attention Blocks	(6, 12, (1, 2, 3))
	Stage 4 Dilated Attention Blocks	(2, 24, (1, 2, 3))
	Global Average Pooling	(1)
Classification Head	FC	(288, 2)
Loss function	ρ	0.001
	Interictal sample weight	0.2
Optimizer	Optimizer	Adam
	Learning rate	0.0001
	Epochs	50
	Batch size	128

Note: Depthwise Conv1d's parameters are (in channels, out channels, kernel size, stride, padding, groups). Dilated Attention Blocks' parameters are (number of blocks, number of attention heads, dilation rates).

size) were empirically optimized to balance convergence stability and computational efficiency, as summarized in Table 3. The selection of hyperparameters in this study was based on prior research experience (Li et al., 2023; Memoria et al., 2025) and preliminary experimental results, and was determined through manual tuning. Specifically, we first referred to representative model configurations in the field of seizure prediction to determine key network architecture parameters, including the number of network layers, convolution kernel size, and stride. Subsequently, the remaining training hyperparameters were manually tuned during the training process.

This study employed the leave-one-seizure-out cross-validation (LOSOCV) strategy, in which the training and testing sets are derived from completely distinct seizure events. This design ensures that segments from the same seizure evolution do not concurrently appear in both sets, thereby completely eliminating temporal data leakage. Specifically, in each iteration, one complete seizure event (including its preictal and interictal segments) is designated as the testing set, while the data from all remaining seizure events of that patient are used for training. The detailed procedure is illustrated in Fig. 7. We adopted segment-based and event-based evaluation criteria to comprehensively assess the performance of the proposed seizure prediction model. In the segment-based evaluation, four performance metrics were used, namely accuracy, sensitivity, specificity, and F1-score. All results are presented as mean ± standard deviation. The formulas for these metrics are as follows:

$$Sensitivity = Recall = \frac{TP}{TP + FN} \times 100\% \tag{13}$$

$$Specificity = \frac{TN}{TN + FP} \times 100\% \tag{14}$$

$$Accuracy = \frac{TP + TN}{TP + FN + TN + FP} \times 100\% \tag{15}$$

$$Precision = \frac{TP}{TP + FP} \times 100\% \tag{16}$$

$$F1score = 2 \times \frac{P \times R}{P + R} \times 100\% \tag{17}$$

where TP (True Positive) denotes the number of correctly predicted preictal segments, TN (True Negative) indicates the number of correctly identified interictal segments, FP (False Positive) refers to the number of interictal segments misclassified as preictal, and FN (False Negative) represents the number of preictal segments misclassified as interictal. Besides, P and R denote precision and recall, respectively.

In the event-based evaluation, it is essential to define the Seizure Prediction Horizon (SPH) and the Seizure Occurrence Period (SOP). As shown in Fig. 8, the SOP represents the time interval during which a seizure is expected to occur, while the SPH refers to the period between the triggering of a prediction alarm and the start of the SOP. This interval provides a critical time window for implementing clinical interventions. A seizure prediction is considered successful if the seizure onset falls within the SOP; otherwise, it is deemed a failure. In this study, the SPH was set to 3 minutes and the SOP to 30 minutes. To reduce the false alarm rate, an early warning signal is triggered only when the system continuously outputs more than 18 positive labels. The event-based evaluation primarily uses sensitivity and the false prediction rate (FPR) per hour to assess system performance.

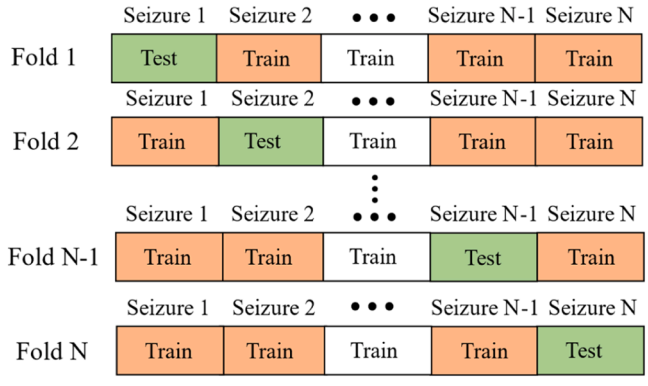


Fig. 7. Schematic diagram of the leave-one-seizure-out cross-validation strategy.

4.2. Experiment results on CHB-MIT database

Table 4 presents the segment-based and event-based evaluation results of the model on the CHB-MIT database. As shown in the table, all subjects achieved more than 91% in accuracy, sensitivity, specificity, and F1-score. The overall average metrics were as follows: accuracy 97.56%, sensitivity 97.75%, specificity 97.50%, and F1-score 97.63%. Notably, Patients 9 and 14 attained perfect scores of 100% across all evaluation metrics. Event-based evaluation revealed an overall average sensitivity of 97.69% and an average false prediction rate of only 0.0232/h. Out of a total of 106 seizure events, the model successfully predicted 102, with only 4 missed predictions occurring exclusively in patients 12 and 15. These results demonstrate that the proposed model exhibits excellent predictive performance and robustness at both the segment and event levels.

4.3. Experiment results on SH-SDU database

Table 5 presents the segment-based and event-based evaluation results of the proposed model on the SH-SDU database. At the segment level, the model achieved an accuracy of 96.93 %, a sensitivity of 97.41%, and a specificity of 95.82%. At the event level, the model attained an average sensitivity of 96.05% and an FPR of 0.0395/h. With the exception of three missed seizure events from three patients, all other seizure events were accurately predicted. Notably, CHB-MIT uses 18 bipolar channels, whereas SH-SDU uses 16 channels after excluding A1 and A2. The stable performance under these two different channel configurations provides experimental support for the adaptability of DCDA-Net to different EEG channel settings after appropriate configuration and training.

5. Discussion

This study constructed a novel end-to-end prediction framework named DCDA-Net by innovatively integrating depthwise convolution and dilated attention mechanism into the seizure prediction task. The depthwise convolution module processes each input channel through independent convolutional operations, effectively maintaining the isolation of channel information and providing a technical basis for accurately identifying key EEG channels associated with epileptic onsets. The dilated attention mechanism combines the expanded receptive field characteristics of dilated convolution with the global dependency modeling capability of self-attention, significantly enhancing the capture of long-range multi-scale temporal features while maintaining local computational efficiency.

Furthermore, because the depthwise convolution performs independent modeling for each channel and the attention mechanism enables adaptive cross-channel feature fusion, the proposed DCDA-Net can effectively accommodate different EEG channel configurations while maintaining stable feature learning capability and prediction performance across databases with different numbers of input channels. Specifically, in this study, the CHB-MIT database uses 18 EEG channels as model inputs, whereas the SH-SDU database uses 16 EEG channels. Despite these differences in channel configurations, DCDA-Net achieves consistently strong and stable prediction performance on both

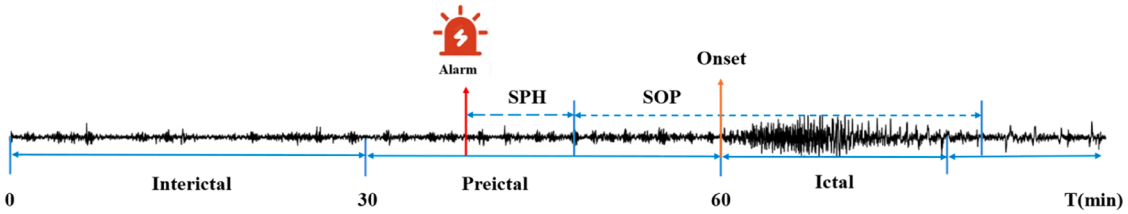


Fig. 8. The definitions of SPH and SOP, and examples of correctly predicting seizure events.

Table 4
The result on CHB-MIT database.

Patient	Selected seizures	Predicted seizures	Segment-based				Event-based	
			Accuracy (%)	Sensitivity (%)	Specificity (%)	F1 Score (%)	Sensitivity (%)	FPR (/h)
1	7	7	99.68±0.11	99.65±0.19	99.72±0.08	99.69±0.11	100.00±0.00	0.00±0.00
2	3	3	98.40±1.15	99.36±0.78	97.41±2.01	98.43±1.14	100.00±0.00	0.00±0.00
3	6	6	98.57±0.98	98.73±1.62	98.41±2.50	98.58±0.95	100.00±0.00	0.00±0.00
4	4	4	98.73±0.82	99.16±0.99	98.31±1.61	98.74±0.81	100.00±0.00	0.00±0.00
5	5	5	98.15±0.05	97.69±2.03	98.92±1.86	98.14±0.24	100.00±0.00	0.00±0.00
6	9	9	93.01±1.04	93.22±0.95	93.83±2.04	93.48±1.10	100.00±0.00	0.00±0.00
7	3	3	99.05±0.36	99.44±0.25	98.47±0.93	99.09±0.40	100.00±0.00	0.00±0.00
8	5	5	99.33±0.23	99.70±0.51	98.96±0.93	99.37±0.22	100.00±0.00	0.00±0.00
9	4	4	100.00±0.00	100.00±0.00	100.00±0.00	100.00±0.00	100.00±0.00	0.00±0.00
10	6	6	94.85±1.01	94.68±2.00	94.92±1.16	94.58±1.07	100.00±0.00	0.00±0.00
12	7	5	95.07±2.01	94.76±2.04	95.85±2.08	95.12±2.16	71.43±48.80	0.29±0.49
13	4	4	97.78±0.91	98.04±0.87	97.34±0.56	97.68±0.98	100.00±0.00	0.00±0.00
14	6	6	100.00±0.00	100.00±0.00	100.00±0.00	100.00±0.00	100.00±0.00	0.00±0.00
15	9	7	95.99±1.41	97.17±1.14	94.68±1.02	96.23±1.28	77.78±44.10	0.22±0.44
16	5	5	94.12±0.22	94.02±0.24	94.43±0.18	94.18±0.32	100.00±0.00	0.00±0.00
17	3	3	99.85±0.28	100.00±0.00	99.74±0.17	99.85±0.25	100.00±0.00	0.00±0.00
18	4	4	95.12±0.99	94.97±1.58	95.39±1.54	95.15±1.01	100.00±0.00	0.00±0.00
19	2	2	99.87±0.51	99.91±0.17	99.77±0.20	99.88±0.45	100.00±0.00	0.00±0.00
20	4	4	96.87±0.72	97.18±0.86	96.63±1.12	96.76±0.62	100.00±0.00	0.00±0.00
21	4	4	99.52±0.28	99.74±0.31	99.05±0.29	99.58±0.27	100.00±0.00	0.00±0.00
22	2	2	95.02±1.37	95.21±1.42	95.81±1.17	95.88±1.42	100.00±0.00	0.00±0.00
23	4	4	97.68±0.77	97.77±0.52	97.29±0.51	97.51±0.67	100.00±0.00	0.00±0.00
Average	-	-	97.56±2.19	97.75±2.26	97.50±2.01	97.63±2.12	97.69±7.54	0.0232±0.08

Table 5
The result on SH-SDU database.

Patient	Selected seizures	Predicted seizures	Segment-based				Event-based	
			Accuracy (%)	Sensitivity (%)	Specificity (%)	F1 Score (%)	Sensitivity (%)	FPR (/h)
1	5	5	98.72±0.21	99.84±0.10	97.40±0.39	98.80±0.18	100.00±0.00	0.00±0.00
2	6	6	98.68±0.26	97.52±0.37	99.82±0.08	98.71±0.18	100.00±0.00	0.00±0.00
3	5	5	94.50±0.38	100.00±0.00	88.99±0.57	94.87±0.42	100.00±0.00	0.00±0.00
4	5	5	96.08±0.18	92.58±0.40	100.00±0.00	96.12±0.20	100.00±0.00	0.00±0.00
5	2	2	100.00±0.00	100.00±0.00	100.00±0.00	100.00±0.00	100.00±0.00	0.00±0.00
6	3	3	98.08±0.29	97.98±0.25	98.42±0.27	98.16±0.22	100.00±0.00	0.00±0.00
7	2	2	96.27±0.39	100.00±0.00	92.37±0.29	96.35±0.27	100.00±0.00	0.00±0.00
8	6	5	96.41±0.28	98.75±0.22	94.01±0.38	96.50±0.32	83.33±40.82	0.17±0.41
9	4	4	92.99±1.88	95.56±1.20	91.22±2.58	93.12±2.01	100.00±0.00	0.00±0.00
10	3	3	94.02±1.11	88.58±3.78	99.18±0.10	93.89±1.51	100.00±0.00	0.00±0.00
11	6	6	98.29±0.98	96.71±0.87	100.00±0.00	98.25±1.01	100.00±0.00	0.00±0.00
12	2	2	98.55±0.52	98.90±0.37	98.20±0.51	98.51±0.46	100.00±0.00	0.00±0.00
13	2	2	100.00±0.00	100.00±0.00	100.00±0.00	100.00±0.00	100.00±0.00	0.00±0.00
14	3	3	99.63±0.64	100.00±0.00	99.26±1.28	99.63±0.63	100.00±0.00	0.00±0.00
15	6	6	99.72±0.56	100.00±0.00	99.44±1.11	99.73±0.55	100.00±0.00	0.00±0.00
16	3	3	97.22±0.79	96.67±4.71	97.78±3.14	97.19±0.90	100.00±0.00	0.00±0.00
17	3	2	90.37±2.95	92.63±7.96	81.11±5.04	83.78±5.24	66.67±57.74	0.33±0.58
18	4	3	92.22±2.14	95.11±1.57	83.33±3.71	84.91±2.35	75.00±50.00	0.25±0.50
19	2	2	100.00±0.00	100.00±0.00	100.00±0.00	100.00±0.00	100.00±0.00	0.00±0.00
Average	-	-	96.93±5.56	97.41±3.81	95.82±8.62	96.24±4.67	96.05±9.77	0.0395±0.10

databases, further demonstrating its adaptability to different channel configurations.

5.1. Ablation experiments

Due to the short intervals between adjacent seizure events in the CHB-MIT database, some preictal EEG segments do not meet the required duration of 30 minutes. Inspired by previous studies (Dong et al., 2024b; Li et al., 2026), we explored a data augmentation strategy based on random splitting and recombination to include more eligible seizure events. Specifically, for EEG segments that fail to satisfy the complete 30-minute preictal window, the available preictal signals

from the same seizure event were randomly partitioned and recombined while preserving local temporal continuity as much as possible, thereby constructing standardized 30-min preictal samples to ensure temporal-scale consistency across different seizure events. This strategy was strictly confined to the available preictal segments within the same seizure event and was mainly used to compensate for missing preictal data ranging from several seconds to several minutes, allowing more eligible seizure events to be included under limited data conditions. Since this strategy was primarily intended to supplement samples with insufficient preictal duration, its actual contribution was further evaluated through an ablation experiment. Accordingly, we compared model performance with and without data augmentation in Table 6. As shown in Table 6, although the data augmentation increased the number of usable seizure events by 21, the improvements across all evaluation metrics were rather limited: accuracy, sensitivity, and event-based sensitivity increased by only 0.11%, 0.13%, and 0.38%, respectively, while specificity decreased slightly. The paired Wilcoxon signed-rank test showed no statistically significant differences between the results obtained with and without data augmentation for any evaluation metric (all $p > 0.05$). The standard deviations of most segment-based metrics also increased after data augmentation, indicating that this strategy did not provide stable performance gains. These results suggest that data augmentation mainly served to supplement usable seizure events, while providing limited improvements in model performance and stability. Therefore, to ensure reproducibility and fair comparison, the results obtained without data augmentation were adopted as the performance outcomes of our proposed approach.

We further evaluated the impact of different convolutional front-end structures on model performance. In this ablation study, Standard CNN, Depthwise Separable CNN, and Depthwise CNN were used and the same back-end dilated attention mechanism and training settings were maintained. As shown in Table 7, Depthwise CNN outperforms the other two structures across most metrics. This suggests that using pure Depthwise CNN as the front end in the DCDA-Net is more conducive to preserving channel-specific representations, allowing for the subsequent dilated attention module to better capture the EEG inter-channel interactions. In contrast, Depthwise Separable CNN introduces cross-channel fusion through pointwise convolution, causing channel information to become prematurely coupled. This early coupling negatively affects the ability of the subsequent module to learn more effective inter-channel interactions.

5.2. Quantitative analysis of seizure onsets

EEG is inherently a multi-channel physiological signal with distinct spatial attributes, where each recording channel corresponds to specific functional brain regions through well-established anatomical relationships. However, most existing deep learning models for seizure detection and prediction treat multi-channel EEG signals as a unified whole during feature extraction, lacking fine-grained analysis of individual channel contributions. This limitation hinders effective channel-level localization, thereby restricting the ability of current methods to provide clinically meaningful spatial information about epileptogenic zones. To address this challenge, this study innovatively employed channel-wise independent convolution operations to preserve the distinctiveness of each channel's features, while evaluating the relative contribution of each channel to classification decisions through the L2 norm of the weight matrices. This approach ultimately enabled quantitative identification and spatial localization of key channels, providing insights on epileptogenic evolution.

Specifically, the DCDA-Net model first performs independent feature extraction on 18 EEG channels via depthwise convolutional layers, thereby maintaining the isolation of information across channels. Subsequently, the contribution of each channel to classification decisions is quantified by calculating the L2 norm of the corresponding convolutional kernels, generating channel importance scores. Finally, these importance scores are mapped into a standard EEG spatial coordinate system according to the 10-20 system to achieve channel localization. Fig. 9 presents a comparative visualization of the model's localization results and the actual seizure regions. Fig. 9 (c) displays a bar chart illustrating the importance distribution of each channel based on L2 norm values. We selected the three key channels with the highest L2 norms, corresponding to the electrode positions highlighted in red in Fig. 9 (a). To further validate their physiological relevance, Fig. 9 (b) shows the raw EEG signals from these three channels during the same time period. It can be observed that these channels already exhibit significant abnormal discharge activity during the pre-seizure phase, while during the seizure phase, the signals display numerous typical spike and sharp wave components, which align closely with the clinical EEG manifestations of epileptic seizures. The results demonstrate a significant spatial correspondence between the high-importance channels identified by the model and the actual epileptogenic zones, validating both the effectiveness and physiological plausibility of the proposed channel analysis method.

Table 8 summarizes the clinically identified seizure onset regions in the CHB-MIT database, the brain regions predicted by the proposed model, and the three key channels with the highest L2 norms for each patient. Specifically, the model correctly identified the seizure onset zone in 15 out of 22 patients, while the predicted regions for most patients showed varying degrees of overlap with the clinically annotated regions. Only Patient 16 was incorrectly localized. Moreover, the three key channels selected by the proposed method were predominantly distributed within or adjacent to the clinically identified seizure onset regions. For example, temporal lobe epilepsy patients (e.g., Patients 1, 3, 7, 8, 14, 17, and 21) mainly exhibited high-L2-norm channels located in temporal-related bipolar leads (e.g., F7–T7, F8–T8, T7–P7, and T8–P8), whereas frontal lobe epilepsy patients (e.g., Patients 2, 5, 9, 12, and 19) showed dominant channels concentrated in frontal-related leads (e.g., FP1–F7, FP1–F3, FP2–F4, and FP2–F8).

5.3. Interpretability analysis of frequency-band responses

Deep learning models are often regarded as "black box" systems, lacking sufficient interpretability in their decision-making processes. To gain deeper insights into the internal working principle of the proposed DCDA-Net model, we employed Grad-CAM technology to visualize the output of the final layer in the depthwise convolution module. As shown in Fig. 10 (a), the visualization results clearly illustrate the model's attention intensity and spatial distribution characteristics across different EEG frequency bands (δ, θ, α, β, γ), providing an intuitive basis for understanding its decision-making logic.

The analysis reveals significant differences in the model's activation patterns across frequency bands, reflecting a hierarchical attention mechanism for multi-band EEG features. In the δ band, the model

Table 6
Comparison of results with and without data augmentation on CHB-MIT.

Method	Segment-based				Event-based	
	Accuracy (%)	Sensitivity (%)	Specificity (%)	F1 Score (%)	Sensitivity (%)	FPR (/h)
Without Data Augmentation	97.56±2.19	97.75±2.26	97.50±2.01	97.63±2.12	97.69±7.54	0.0232±0.08
Data Augmentation	97.67±2.32	97.88±2.37	97.48±2.26	97.71±2.26	98.07±6.40	0.0183±0.06

Table 7
Comparison of results for different convolution methods on CHB-MIT.

Method	Segment-based				Event-based	
	Accuracy (%)	Sensitivity (%)	Specificity (%)	F1Score (%)	Sensitivity (%)	FPR (/h)
Standard CNN	95.73±2.88	95.30±3.12	95.78±2.70	95.75±2.81	95.62±8.72	0.0346±0.09
Depthwise Separable CNN	97.41±2.23	97.17±2.45	97.46±2.14	97.55±2.30	96.38±8.31	0.0269±0.08
Depthwise CNN	97.56±2.19	97.75±2.26	97.50±2.01	97.63±2.12	97.69±7.54	0.0232±0.08

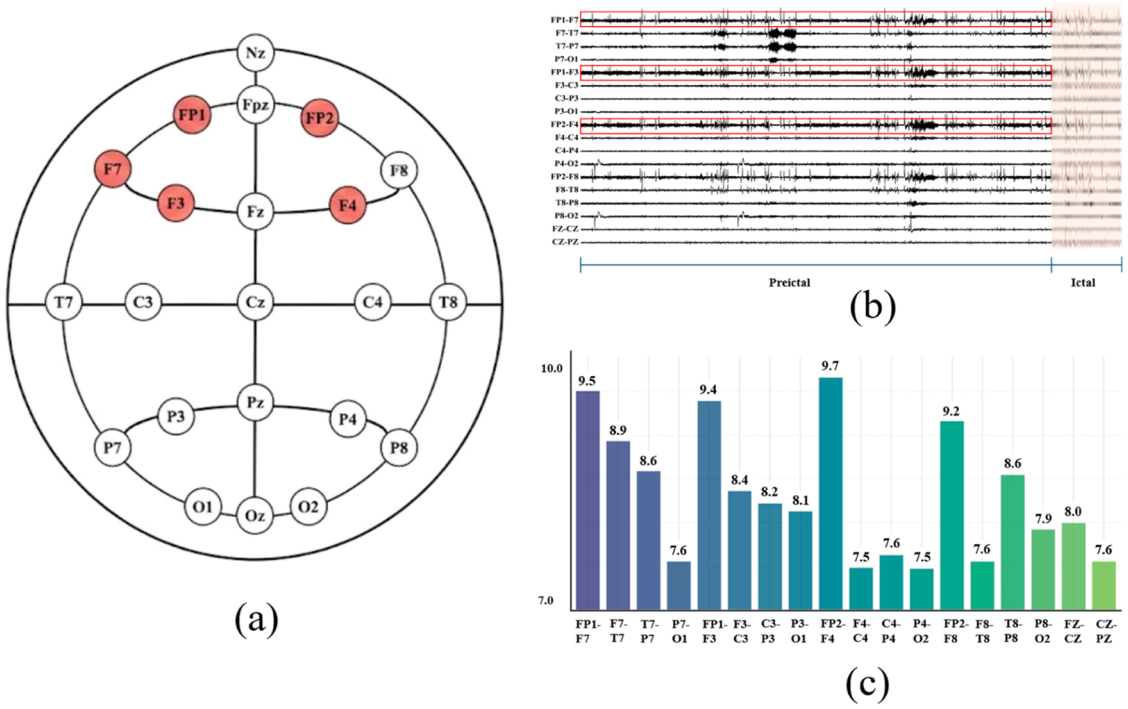


Fig. 9. Visualization of model localization results compared with actual seizure regions. (a) Electrode positions. (b) EEG during the seizure period. (c) L2 norm values across different channels.

Table 8
Seizure onset regions in the CHB-MIT database and the three key channels with the highest L2 norms.

Patient	Seizure area	Predicted area	Three Key Channels
1	T	T	F7-T7, F8-T8, T8-P8
2	F	F	FP1-F7, FP1-F3, FP2-F4
3	T	T	T7-P7, F8-T8, T8-P8
4	T, O	T, O	F8-T8, T8-P8, P3-O1
5	F	F	FP1-F7, FP1-F3, FP2-F8
6	T	F, T	F7-P7, F3-C3, F8-T8
7	T	T	F7-T7, T7-P7, T8-P8
8	T	T	T7-P7, F8-T8, T8-P8
9	F	F	F3-C3, FP2-F8, F4-C4
10	T	T, O	F8-T8, T8-P8, P4-O2
12	F	F	FP1-F7, FP1-F3, FP2-F4
13	T, O	F, T	F7-T7, F8-T8, F8-T8
14	T	T	F7-T7, T7-P7, F8-T8
15	F, T	F	FP1-F7, FP2-F4, FP1-F3
16	T	F	FP1-F3, F3-C3, FZ-CZ
17	T	T	T7-P7, F8-T8, T8-P8
18	T, O	F, O	FP2-F8, P3-O1, P4-O2
19	F	F	FP1-F3, FP2-F4, FP2-F8
20	T	F, T	T8-P8, F4-C4, F8-T8
21	T	T	F7-T7, F8-T8, T8-P8
22	T, O	T, O	T7-P7, F8-T8, P4-O2
23	F	F	FP1-F3, FP1-F7, FP2-F8

primarily focuses on low-frequency, slowly varying components. They are typically associated with cortical slow-wave activity and global synchronization phenomena, possibly reflecting the overall modulation of brain functional state before seizure onset. In the θ band, the model's high-response regions concentrate on rhythmic oscillation segments, indicating its ability to effectively capture mid-to-low frequency characteristics related to pre-seizure cognitive fluctuations and changes in cortical excitability. In the α band, the Grad-CAM heatmap shows that the model pays particular attention to intervals with stable rhythms but significant local energy fluctuations, suggesting a potential link to decreased interregional synchronization and rhythm disruption during the pre-seizure phase. In the β and γ bands, the model's activations exhibit localized enhancement features, demonstrating high sensitivity to high-frequency burst activities and motor-related rhythms. Fig. 10 (b) illustrates the L2 norm weights across different frequency bands after model training. It can be observed that during the identification of the preictal period, the model primarily focuses on the high-frequency γ band. In contrast, the model places greater emphasis on the mid-frequency β rhythm when distinguishing the baseline interictal state. The Grad-CAM visualization results demonstrate that the DCDA-Net model exhibits hierarchical and frequency-specific attention characteristics, revealing its interpretable feature extraction mechanism in EEG-based seizure prediction tasks.

5.4. Feature visualization

Furthermore, we employed t-SNE to visualize the distribution of

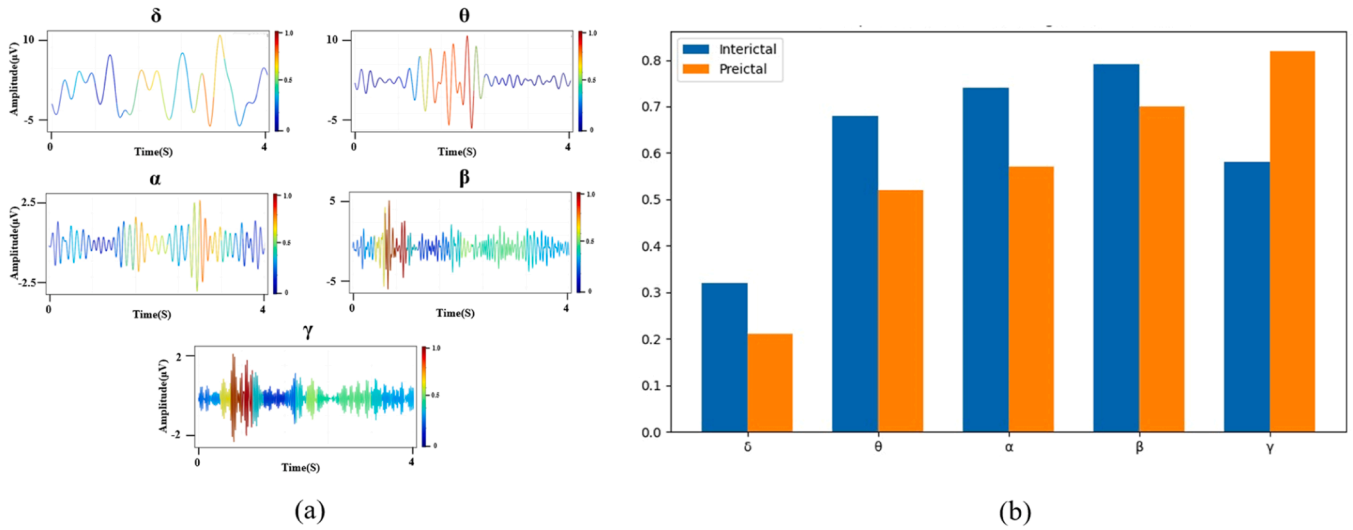


Fig. 10. Model's focus on different frequency bands. (a) Grad-CAM results for the depthwise convolution module. δ, θ, α, β, γ denote different EEG frequency bands. (b) The L2 norm weights for different bands after model training.

features during the feature extraction process, as shown in Fig. 11. It can be observed that the original EEG signals exhibit mixed feature distributions with no clear separability between different classes. After processing by the depthwise convolution module, the features of the two classes begin to show clustering tendencies, demonstrating preliminary separation characteristics. Following further processing by the dilated attention module, the features of different classes form clearer and more distinct cluster distributions in the embedding space. These visualization results further validate the effectiveness and interpretability of the model in distinguishing between pre-seizure and interictal EEG patterns from the perspective of feature learning.

5.5. Effects of temporal weighted loss

To evaluate the performance improvement brought by the proposed temporal weighted loss, we examined the effect of replacing it with Cross Entropy Loss while keeping all other hyperparameters unchanged. The results are shown in Table 9. Fig. 12 depicts the differences in accuracy and sensitivity among different patients. It can be clearly observed that the model trained with Temporal Weighted Loss outperforms the one trained with Cross Entropy Loss in both segment-based and event-based evaluation criteria. In the segment-based evaluation, the model shows improvements of 1.61% in accuracy, 1.88% in sensitivity, and 1.42% in specificity. In the event-based evaluation, the model achieves a 0.55% improvement in sensitivity. These results confirm that the temporal weighted loss contributes to enhanced effectiveness in seizure prediction.

We systematically evaluated the impact of the decay coefficient ρ in the temporal weighted loss on the model performance. By setting ρ to five different values (0.0001, 0.0005, 0.001, 0.0025, and 0.005) and training five model groups under identical parameters, we obtained the comparative results shown in Fig. 13. Fig. 13 (a) displays the weight distribution range corresponding to each ρ value, with the maximum weight reaching 0.5. Fig. 13 (b) presents the ROC curves of the models trained with different ρ values on the same test set. It shows that the model achieves optimal performance at ρ=0.001 (AUC=0.9936), while performing worst at ρ=0.005 (AUC=0.7574). Fig. 13 (c) illustrates the average performance across all patients tested for different ρ values. It can be observed that when $\rho \leq 0.001$, model performance improves with increasing ρ values, peaking at ρ=0.001. However, when $\rho > 0.001$, model performance declines significantly, potentially due to excessive weight variation amplitude leading to unstable optimization.

5.6. Comparison with existing methods

Table 10 summarizes and compares the seizure prediction performance of representative studies on the CHB-MIT database. All the compared methods adopt patient-specific modeling, which is consistent with this study. Although differences exist in preprocessing procedures and experimental settings, the comparison still provides a meaningful reference at the overall performance level.

Khan et al. (2018) employed continuous wavelet transform (CWT) to extract time-frequency features and utilized CNNs for classification. Their method achieved an event-based sensitivity of 87.80% with an

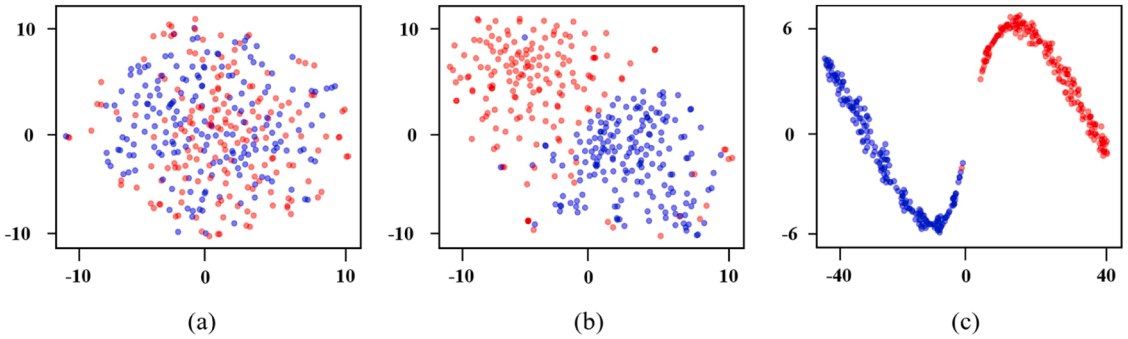


Fig. 11. Visualization of t-SNE of the extracted features. (a) The original EEG data. (b) The features extracted by the depthwise convolution module. (c) The features extracted by the dilated attention module.

Table 9
Comparison of results for cross entropy loss and temporal weighted loss on CHB-MIT.

Method	Segment-based				Event-based	
	Accuracy (%)	Sensitivity (%)	Specificity (%)	F1 Score (%)	Sensitivity (%)	FPR (/h)
Cross Entropy Loss	95.95±2.86	95.87±2.77	96.08±2.57	95.99±2.84	96.78±8.40	0.0323±0.09
Temporal Weighted Loss	97.56±2.19	97.75±2.26	97.50±2.01	97.63±2.12	97.69±7.54	0.0232±0.08

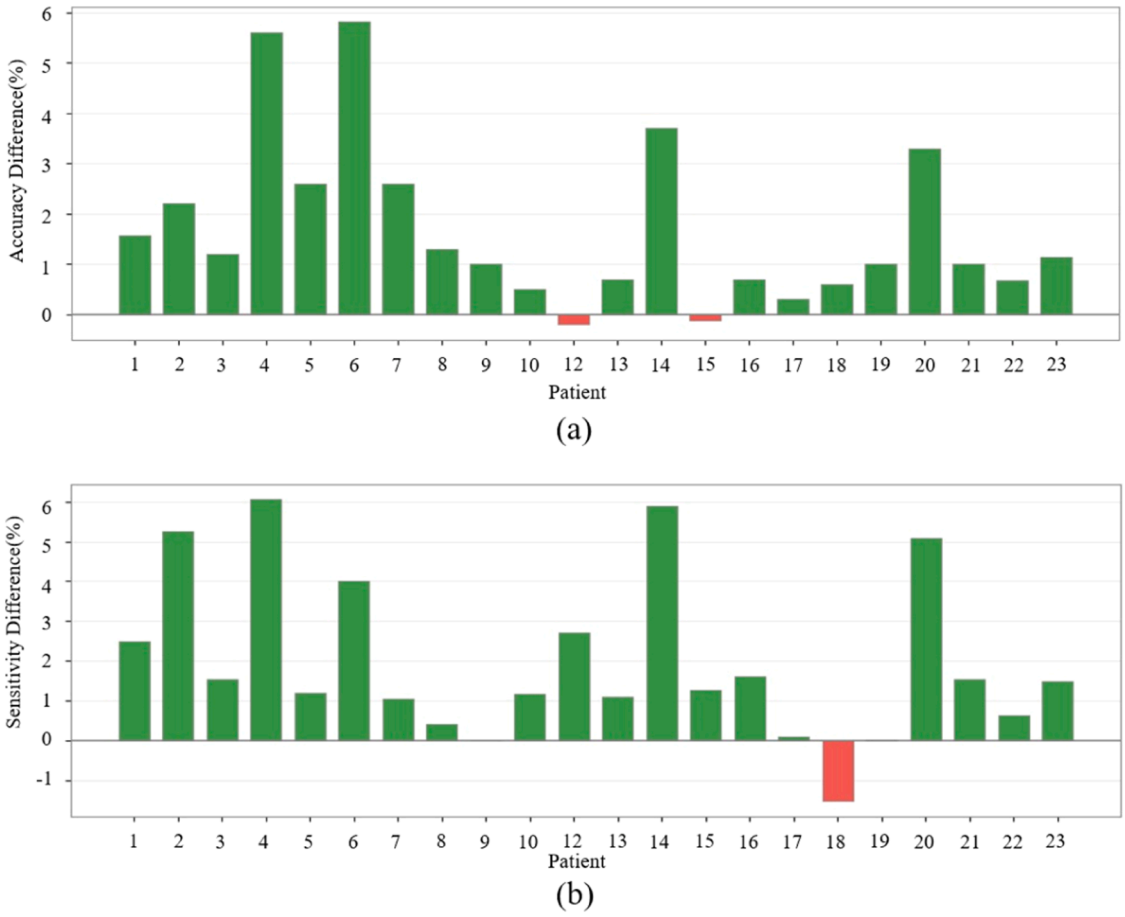


Fig. 12. Differences in accuracy and sensitivity between patients. (a) differences in accuracy, (b) differences in sensitivity.

FPR of 0.1470/h, indicating that there remains room for performance improvement. Similarly, Truong et al. (2018) proposed a prediction model consisting of three convolutional blocks and used short-time Fourier transform (STFT)-based time-frequency spectrograms as input, achieving an event-based sensitivity of 81.20% and an FPR of 0.1600/h, both of which are inferior to the results obtained by our method. From a feature-engineering perspective, Hellar et al. (2022) extracted key spectral features using embedded dynamic mode decomposition (EmDMD) and performed seizure prediction with the RF classifier. Compared with our approach, their reported segment-based sensitivity (92.80%), specificity (89.70%), and F1-score (91.40%) are relatively lower. Ding et al. (2023) incorporated a multi-head attention (MHA) mechanism into the backend of a multi-layer CNN, which increased the event-based sensitivity to 99.45%; however, the segment-based accuracy (86.64%), sensitivity (86.45%), specificity (86.45%), and F1-score (86.80%) all decreased, revealing an imbalance between event-based and segment-based performance. To effectively model the time-frequency characteristics of EEG signals, the DenseNet-ViT method (Yuan et al., 2024) integrated the advantages of local and global feature extraction to learn STFT-based time-frequency representations, achieving a segment-based accuracy of 93.65%. The end-to-end LSTM model (Hasan et al., 2025) explicitly captured long-range temporal dependencies in EEG signals and achieved relatively balanced performance. In comparison, the proposed method yields an overall performance improvement of nearly 2% while additionally providing clinically relevant event-based analysis. Zhai et al. (2025) proposed a lightweight seizure prediction model that integrates a residual network with a kernel-enhanced global temporal attention block (K-GTA Block), achieving a segment-based accuracy of 93.94% and an event-based sensitivity of 100% on the CHB-MIT database. The multi-domain feature hybrid (MFH) model (Ji et al., 2026) further incorporated EEG source localization and graph-theoretical analysis to interpret brain functional mechanisms underlying seizure prediction; nevertheless, its accuracy (96.82%), sensitivity (96.35%), specificity (97.19%), and F1-score (96.53%) remain slightly lower than those of our method. Qu et al. (2026) proposed a causal attention network (CANet) that fuses time-frequency channel features, achieving a segment-based accuracy of 95.09% and an event-based sensitivity of 97.06% on the CHB-MIT database.

Compared with the aforementioned approaches, the proposed DCDA-Net model demonstrates superior and more balanced performance under both segment-based and event-based evaluation criteria,

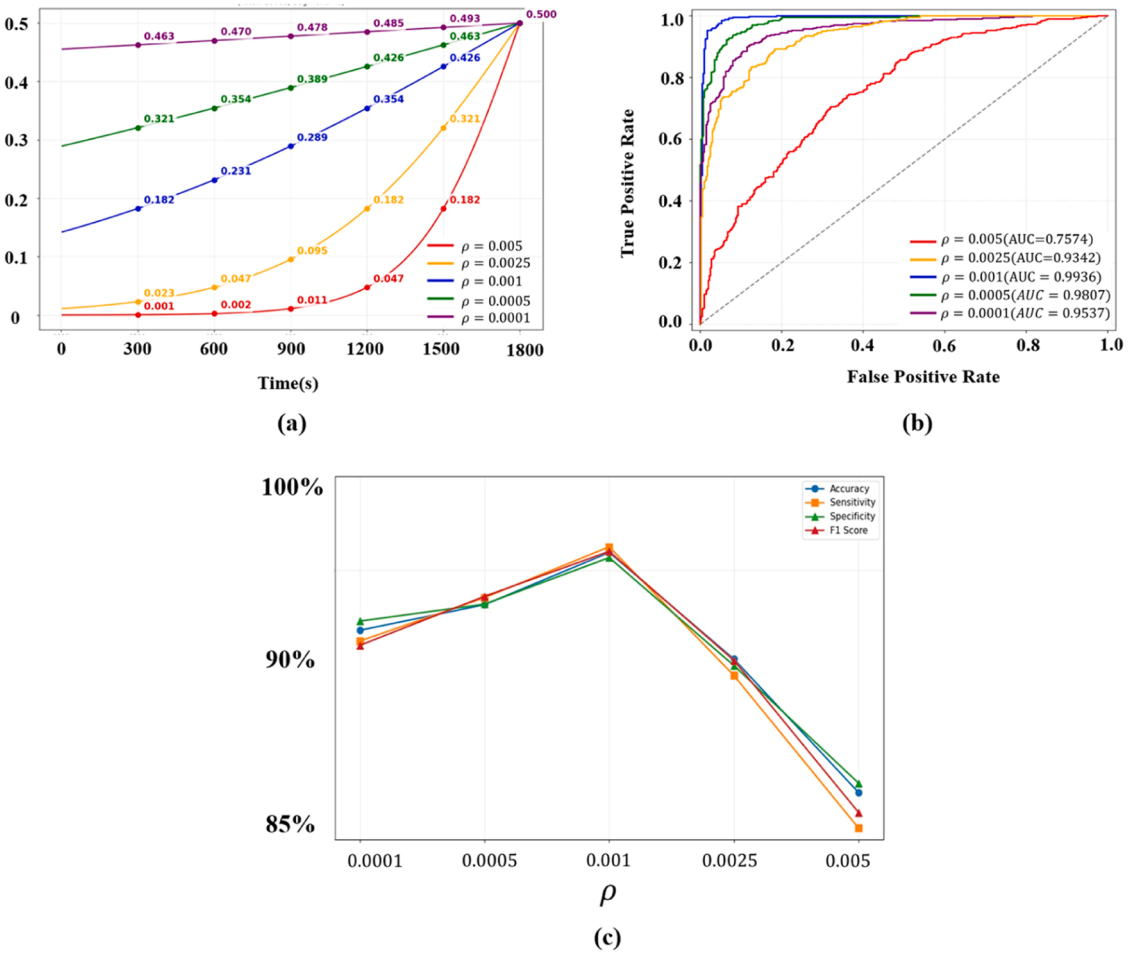


Fig. 13. Results of ablation experiments with different ρ settings. (a) The curve of the temporal weight loss value over time. (b) A comparison of ROC curves for models corresponding to different temporal decay coefficients ρ. (c) The test results for different temporal decay coefficients ρ.

Table 10
Comparison of the performance for different methods on CHB-MIT database.

Authors	Method	Segment-based				Event-based	
		Accuracy (%)	Sensitivity (%)	Specificity (%)	F1 Score (%)	Sensitivity (%)	FPR (/h)
Khan et al. (2018)	CWT+CNN	-	-	-	-	87.80	0.1470
Truong et al. (2018)	STFT+CNN	-	-	-	-	81.20	0.1600
Hellar et al. (2022)	EmDMD+RF	-	92.80	89.70	91.40	-	-
Ding et al. (2023)	CNN+MHA	86.64	86.81	86.45	86.80	99.45	0.0348
Yuan et al. (2024)	STFT+ DenseNet-ViT	93.65	93.56	-	93.40	-	0.0840
Hasan et al. (2025)	LSTM	95.37	95.80	-	95.00	-	-
Zhai et al. (2025)	Lightweight GTA	93.94	94.00	93.91	-	100.00	0.0755
Ji et al. (2026)	CWT+MFH	96.82	96.35	97.19	96.53	-	0.02696
Qu et al. (2026)	CANet	95.09	93.95	96.23	-	97.06	0.0251
This work	DCDA-Net (Data Augmentation)	97.67	97.88	97.48	97.71	98.07	0.0183
	DCDA-Net (Without Data Augmentation)	97.56	97.75	97.50	97.63	97.69	0.0232

validating its effectiveness and advantage for epileptic seizure prediction. Notably, under the setting without data augmentation, DCDA-Net still outperforms existing state-of-the-art methods on multiple metrics, benefiting from the innovative design of dilated attention mechanism and temporal weighted loss function.

5.7. Limitations and future work

Despite the competitive predictive performance achieved in this study, several limitations remain. The current work has demonstrated the physiological plausibility of the identified channel-importance scores through qualitative analysis and visual validation, but the accuracy of channel localization has only been validated in the CHB-MIT database. Because the SH-SDU database does not include annotations of epileptic seizure onset regions, quantitative evaluation of channel localization agreement was not feasible. This framework will employ metrics such as top-k localization accuracy and regional overlap ratio to assess the spatial agreement between the key channels identified by the

model and the clinically annotated seizure regions. In addition, more dynamic definition strategies of the SPH and SOP will be investigated to adapt to individualized seizure patterns, and the computational complexity of the model will be further optimized to enhance its feasibility for real-time deployment. Finally, we will further focus on cross-subject evaluation using patient-independent protocols, such as leave-one-subject-out cross-validation, to provide a more rigorous assessment of the model's generalization.

6. Conclusion

This study proposes an efficient deep learning framework, namely DCDA-Net, tailored for EEG-based seizure prediction. Fundamentally, the innovation of our approach lies in its synergistic, task-specific architecture rather than a mere combination of existing deep learning modules. In the DCDA-Net model, a depthwise CNN module is first employed to perform independent feature extraction from multi-channel EEG signals, effectively preserving spatial information across channels. Then, an innovative dilated attention module is introduced. By uniquely fusing the exponentially expanded receptive field of dilated convolutions with the global dependency modeling of self-attention, it efficiently captures complex, multi-scale temporal preictal dynamics while maintaining crucial local contextual correlation. In addition, the pioneering introduction of a temporal weighted loss function shifts the paradigm of seizure prediction from uniform sample optimization to temporally aware learning, dynamically guiding the model to focus on the time windows most critical to prediction. The proposed framework is comprehensively evaluated on the public CHB-MIT database and the clinical SH-SDU database, achieving leading performance across multiple segment-based and event-based evaluation metrics. Moreover, by quantitatively analyzing the contributions of individual EEG channels, the model successfully identifies key channels that are highly correlated with epileptogenic pathways, offering valuable insights into seizure initiation and spread.

CRediT authorship contribution statement

Xingchen Dong: Writing – original draft, Visualization, Validation, Software, Methodology, Data curation, Conceptualization. **Haotian Li:** Writing – original draft, Software, Formal analysis, Data curation, Conceptualization. **Qiangqiang Chen:** Visualization, Formal analysis, Data curation. **Xiaoyu Chen:** Formal analysis, Data curation. **Weisen Lu:** Visualization, Validation. **Landi He:** Validation, Software. **Chung Tin:** Writing – review & editing, Validation, Supervision. **Weidong Zhou:** Writing – review & editing, Supervision, Methodology, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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